09 | Tensor-valued cuts: endpoint versus path conductance

Fixed scalar target F; inner products sum over every coordinate in component z_k.

Endpoint component

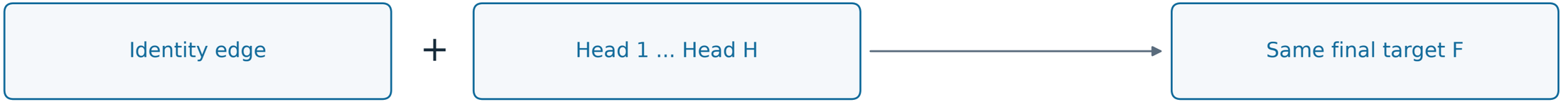
$$E^C_{k,i} = x_i \langle \nabla_{z_k} F(x), \partial_i z_k(x) \rangle$$

$$\sum_k E^C_{k,i} = x_i \partial_i F(x)$$

Input-resolved path conductance

$$K^C_{k,i} = \Delta x_i \int_0^1 \langle \nabla_{z_k} F(x_\alpha), \partial_i z_k(x_\alpha) \rangle d\alpha$$

$$\sum_k K^C_{k,i} = IG_i(F; x, x_0)$$



Each input-to-F path crosses exactly one registered edge. Heads include the native projection and downstream graph.

Residual/head edge cuts and attention differential paths are two forms of structured decomposition.

Each block or cut reconstructs its own registered total. Do not add different cuts, or routing together with Q and K.

Closure is a chain-rule property. Finite interventions require their own directions, reference activations and measurements.